

Eighteen pitfalls to beware of in AI journalism

A checklist for avoiding hype



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Reporting about AI is hard. Companies hype their products and most journalists aren't sufficiently familiar with the technology. When new journalists uncritically repeat PR statements, **overuse images of robots**, **attribute to AI tools**, or **downplay their limitations**, they **mislead** and **misinform** about the **potential and limitations of AI**.

We noticed that many articles tend to mislead in similar ways, so we analyzed over 50 articles about AI from major publications, from which we compiled a list of recurring pitfalls. We hope that being familiar with these will help you avoid hype whenever you see it. We also hope this compilation of pitfalls will help journalists avoid them.

We were inspired by many previous efforts at dismantling hype in news reporting on AI by **Prof. Emily Bender**, **Prof. Ben Shneiderman**, **Lakshmi Sivadas** and **Sabrina Argoub**, **Prof. Emily Tucker**, and **Dr. Daniel Leifer**.

You can download a PDF checklist of the 18 pitfalls with examples here.

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Example 1: “The Machine Are Learning, and We are the students” (NYT)

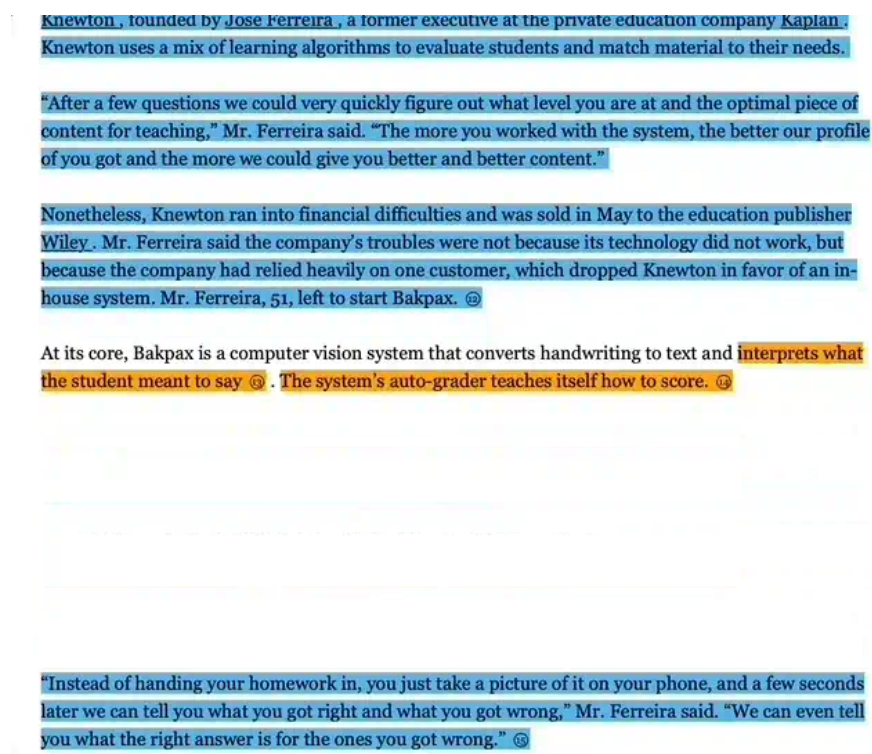
- We identified 19 issues in this article, which you can read [here](#).

In December 2019, NYT published a piece about educational technology (EdTech) product called Bakpax. It is a 1,500-word, feature-length article that provides neither accuracy, balance, nor context.

It is sourced almost entirely from company spokespeople, and the article borrows liberally from Bakpax's PR materials to exaggerate the role of AI. To keep the spotlight on AI, the article downplays the human labor from that keeps the system running—such as developing and digitizing assignments.

Bakpax shut down in May 2022.

This is hardly a surprise: EdTech is an overhyped space. In the last decade there have been hundreds of EdTech products that claim to “revolutionize” education. Despite [billions of dollars in funding](#), many of them [fail](#). Unfortunately, the article does not provide any context about this history.



A GIF of our annotations showing the prevalence of issues throughout the article. Each annotation represents a pitfall. You can read the entire annotated article [here](#)

Example 2: “AI may be as effective as me specialists at diagnosing disease” (CNN)

- We identified 9 issues in this article, which you can read [here](#).

In September 2019, CNN published this article about an AI research buries the lede, seemingly intentionally: the spotlight is on the success tools in diagnosis, whereas the study finds that fewer than 1% of paper tools follow robust reporting practices. In fact, an expert quoted at the end of the article stresses that this is the real message of the study.

Besides, the cover image of the article is a robot arm shaking hands with a human, even though the study is about finding patterns in medical data. These humanoid images give a misleading view of AI, as we’ve described

Example 3: “AI tested as university exam undergo digital shift” (FT)

- You can read the annotated article [here](#).

Many schools and universities have adopted remote proctoring software during the COVID-19 pandemic. These tools suffer from **bias** and **lack of validity**, enable **surveillance**, and raise other concerns. There has been an **uproar against remote proctoring** from **students**, **non-profits**, and even **senators**.

In November 2021, the Financial Times published an article on a proctoring tool called Sciolink that presents an entirely one-sided view of remote proctoring. It almost exclusively quotes the creators of Sciolink and provides no context about the **limitations and risks of remote proctoring tools**.

Eighteen pitfalls in AI journalism

We analyzed over 50 news stories on AI from 5 prominent publications: The New York Times, CNN, Financial Times, TechCrunch, and VentureBeat.

We briefly discuss the 18 pitfalls below; see our **checklist** for more details and examples.

Flawed human-AI comparison

What? A false comparison between AI tools and humans that implies that AI tools and humans are similar in how they learn and perform.

Why is this an issue? Rather than describing AI as a broad set of tools, these comparisons anthropomorphize AI tools and imply that they have the same capabilities as humans.

potential to act as agents in the real world.

Pitfall 1. Attributing agency to AI: Describing AI systems as taking action independent of human supervision or implying that they may soon be able to do so.

Pitfall 2. Suggestive imagery: Images of humanoid robots are often used to illustrate articles about AI, even if the article has nothing to do with robots. This gives readers a false impression that AI tools are embodied, even though AI is just software that learns patterns from data.

Pitfall 3. Comparison with human intelligence: In some cases, articles imply that AI algorithms learn in the same way as humans do. For example, comparisons of deep learning algorithms with the way the human brain performs certain functions are common. Such comparisons can lend credence to claims that AI is “sentient”, as Dr. Timnit Gebru and Dr. Margaret Mitchell note in their [recent op-ed](#).

Pitfall 4. Comparison with human skills: Similarly, articles often compare how well AI tools perform with human skills on a given task. This falsely implies that AI tools and humans compete on an equal footing—hiding the fact that AI tools only work in a narrow range of settings.

Hyperbolic, incorrect, or non-falsifiable claims about AI

What? Claims about AI tools that are speculative, sensational, or incite the spread of hype about AI.

Why is this an issue? Such claims give a false sense of progress in AI and make it difficult to identify where true advances are being made.

Pitfall 5. Hyperbole: Describing AI systems as revolutionary or

groundbreaking without concrete evidence of their performance giving an impression of how useful they will be in a given setting. This issue is when AI tools are deployed in a setting where they are known to have failures—we should be skeptical about the effectiveness of AI tools in these settings.

Pitfall 6. Uncritical comparison with historical transformations: Comparing AI tools with major historical transformations like the invention of electricity or the industrial revolution is a great marketing tactic. However, when news articles adopt these terms, they can convey a false sense of potential progress—especially when these claims are not backed by real-world evidence.

Pitfall 7. Unjustified claims about future progress: Claims about how future developments in AI tools will affect an industry, for instance, by implying that AI tools will inevitably be useful in the industry. When these claims are made without evidence, they are mere speculation.

Pitfall 8. False claims about progress: In some cases, articles include exaggerated or false claims about what an AI tool can do.

Pitfall 9. Incorrect claims about what a study reports: News articles often cite academic studies to substantiate their claims. Unfortunately, there is often a gap between the claims made based on an academic study and what the study reports.

Pitfall 10. Deep-sounding terms for banal actions: As Prof. Emily Bender discusses in her work on dissecting AI hype, using phrases like “the **act of next-word prediction**” or “**the magic of AI**” implies that an AI tool is doing something remarkable. It hides how mundane the tasks are.

Uncritically platforming those with self-interest

What? News articles often use PR statements and quotes from company spokespeople to substantiate their claims without providing adequate context or balance.

Why is this an issue? Emphasizing the opinions of self-interested parties without providing alternative viewpoints can give an over-optimistic view of progress.

Pitfall 11. Treating company spokespeople and researchers as neutral

When an article only or primarily has quotes from company spokespeople or researchers who built an AI tool, it is likely to be over-optimistic about the potential benefits of the tool.

Pitfall 12. Repeating or re-using PR terms and statements: News articles often re-use terms from companies' PR statements instead of describing how an AI tool works. This can misrepresent the actual capabilities of a tool.

Limitations not addressed

What? The potential benefits of an AI tool are emphasized, but the potential limitations are not addressed or emphasized.

Why is this an issue? A one-sided analysis of AI tools can hide the potential limitations of these tools.

Pitfall 13. No discussion of potential limitations: Limitations such as inadequate validation, bias, and potential for dual-use plague most AI tools. When these limitations are not discussed, readers can get a skewed view of the risks associated with AI tools.

Pitfall 14. Limitations de-emphasized: Even if an article discusses limitations and quotes experts who can explain them, limitations are often downplayed by the structure of the article, for instance by positioning them at the end of the article or giving them limited space.

Pitfall 15. Limitations addressed in a “skeptics” framing: Limitations of AI tools can be caveated in the framing of the article by positioning experts who explain these limitations as skeptics who don’t see the true potential of AI. Prof. Bender discusses this issue in **much more detail** in her response to the NYT Mag article.

Pitfall 16. Downplaying human labor: When discussing AI tools, articles often foreground the role of technical advances and downplay all the human labor that is necessary to build the system or keep it running. The book **G** by Dr. Mary L. Gray and Dr. Siddharth Suri reveals how important human labor is. Downplaying human labor misleads readers into thinking that AI tools work autonomously, instead of clarifying that they require **significant overhead** in terms of human labor, as Prof. Sarah T. Roberts discusses.

Pitfall 17. Performance numbers reported without uncertainty estimates and caveats: There is seldom enough space in a news article to explain how performance numbers like accuracy are calculated for a given application and what they represent. Including numbers like “90% accuracy” in the headline of the article without specifying how these numbers are calculated can easily misinform readers about the efficacy of an AI tool. Moreover, AI tools often suffer from performance degradations under even slight changes to the data they are evaluated on. Therefore, absolute performance numbers can be misleading for readers about the efficacy of these tools in the real world.

Pitfall 18. The fallacy of inscrutability: Referring to AI tools as inscrutable

black boxes **is a category error**. Instead of holding the developers of tools accountable for their design choices, it shifts scrutiny to the technical aspects of the system. Journalists should hold developers accountable for the performance of AI tools rather than referring to these tools as black boxes, allowing developers to evade accountability.

While these 18 pitfalls appear in the articles we analyzed, there are broader issues that go beyond individual articles. For instance, when narratives about AI wiping out humanity are widely discussed, they overshadow current world problems such as bias and the lack of validity in AI tools. This underscores that news media must play its **agenda-setting role** responsibly.

We thank Klaudia Jaźwińska, Michael Lemonick, and Karen Rouse for their feedback on a draft of this article. We re-used the code from Molly Whalen's **The Edited Latecomers' Guide to Crypto** to generate our annotated version.



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Leon Galindo Stenutz 30 Sep 2022



Excellent, informative, serious -- and a (temporary?) relief from the onslaught of info dangers of runaway AI.

Two core questions:

1. Is AI truly a core species-level anthropogenic existential risk?
2. How much of what you have observed is merely the result of bad journalism and public relations -- versus a partial or upfront attempt to hype up AI Risk as a form of even intentional PsyOps focused on destabilizing certain audiences and societies?

Would also be interesting to better understand how your insights relate to the alarm as an existential risk sounded by the likes of Stephen Hawking, Elon Musk, Eliezer Y Gates, Stuart Russell.

Also, to understand how your research and vision of the near to mid-term future of AI align with, or differ from the work of Nick Bostrom, Jaan Taallin, Max Tegmark, Yann Christian, or Melanie Mitchell, among others.

Thanks, appreciate your article deeply.

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Abi Olvera 4 Aug 2024

Excellent and useful piece. The very last link for the "agenda setting quality" of media. Were there other citations to replace it?

I can see the "black box" nature of AI being a cop out for applications like those in E. The "black box" discussion be useful in some limited circumstances like general purpose AI.

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